

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **FIVE** questions taking at least two questions from each group]

Group - A

1. **A and B are two events where $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$, and $P(A \cup B) = \frac{10}{12}$. A professor said that $P(A|B) = P(B|A)$.**

- What is the minimum value of probability? 1
- Why can't probability be negative? Explain with an example or theoretically. 2
- Find $P(A \cap B)$ from the stem. 3
- Evaluate the professor's assertion. 4

2. A box contains 5 green, 8 yellow, and 7 orange marbles. A child takes out three balls at random.

- (a) What is a simple event? 1
- (b) Prove using Venn diagram $P(A \cap \bar{B}) = P(A) - P(A \cap B)$ 2
- (c) What is the probability that two marbles are yellow and one green? 3
- (d) Determine the probability that the marbles are the same color. 4

3. The joint probability function of two random variables X and Y is given by:

$$P(X, Y) = \frac{3x + y}{48}; \quad x = 1, 2; \quad y = 0, 1, 2, 3$$

- (a) What is a discrete random variable? 1
- (b) If $a < b$, $F(b) - F(a) = ?$ where F is cumulative distribution function? Derive mathematically. 2
- (c) Find $P(X)$. 3
- (d) Calculate $P(X|Y)$ and $P(X|Y = 1)$. 4

4. A company wants to study the relationship between its advertising expenditure (in thousand \$) and monthly sales (in thousand \$). Data for 6 months are given below:

Advertising (X, \$'000)	2	4	6	8	10	12
Sales (Y, \$'000)	48	56	62	70	78	84

- What is correlation? 1
- Make a scatter plot from the given data and propose the association of the variables. 2
- Find and interpret the correlation coefficient. 3
- Fit the regression line and predict the monthly sales when advertising expenditure is \$9,000. 4

Group - B

5. The first four moments around 2 of a dataset were the following:

-1, 5, 20, 90

- (a) What is raw moment? 1
- (b) What is the standard deviation of the data in the stem? 2
- (c) Determine the third central moment. 3

(d) Comment on the kurtosis of the given data. 4

6. The probability density function of a continuous random variable is

$$f(x) = \begin{cases} kx(x - 1), & 1 \leq x \leq 4 \\ 0, & otherwise \end{cases}$$

- (a) What is the range of probability? 1
- (b) Find the value of k 2
- (c) Find the Expectation of the function. 3
- (d) Find $V(3X+2)$. 4